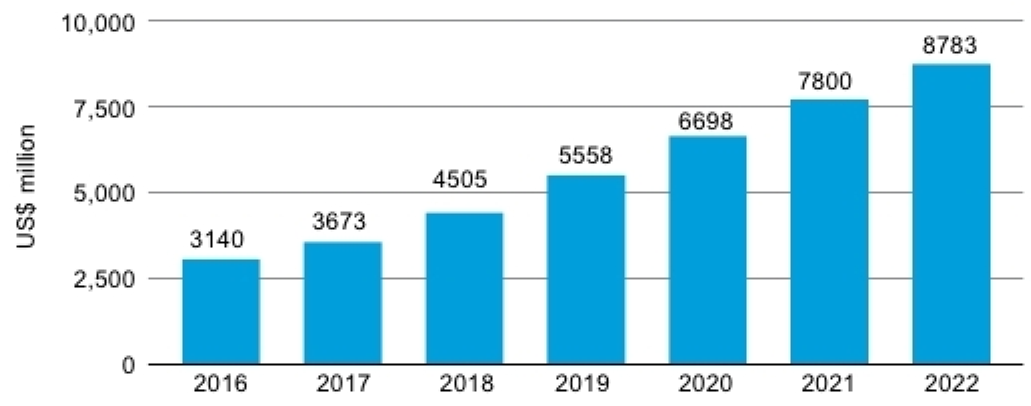
Consumer electronics forecast

According to *www.statista.com*, revenue from consumer electronics in India is expected to show an annual growth rate (CAGR) of 18.2 per cent, in 2018-2022, resulting in a market volume of US$ 8.8 billion by 2022.



Revenue from consumer electronics in India (Credit: www.statista.com)

systems that enable automatically turning on and off the television.

Cellphones. There are multitudes of sensors in modern smartphones. Depending on the brand and model, you can find sensors like GPS, Bluetooth, fingerprint, accelerometer, motion, vibration, sound, gyroscope, magnetometer, gravity, linear acceleration, rotation, ambient temperature, proximity, light, pressure, relative humidity and more.

Smart homes. Smart home technology provides security, comfort, convenience and energy efficiency, by allowing you to control your home through smart devices and, often, a smart home app installed on your smartphone. Sensors in automatic doors, faucets, lightings, entertainment systems and appliances let you to control and monitor these in a smart way.

Smart vending machines. Coffee, tea and milk vending machines need to dispense enough milk or water as per the recipe. Apart from input switches, this is accomplished by filling up a secondary reservoir fitted with float and magnetic sensors. Signals through these sensors can be used to trigger other solenoids or pumps, which stop the main reservoir output and open the drain valve from the secondary reservoir to dispense the liquid into a cup. Smart features of vending machines include providing several cups of coffee with one press, and wireless control through mobile or computer network to control the amount of sugar, water, coffee and others.

Temperature of the indoor environment of the machine is monitored through the network, which adjusts the taste of coffee or tea according to specification. A counter is used to indicate consumption of coffee, tea and milk. The machine and the phone exchange data via Bluetooth. You can find various sensors in smart vending machines, including miniature flat pack, mountable and cylindrical.

Smart security systems. Three main components in a security system are camera, sound sensor (or microphone) and night-vision IR sensor. Security also comes with a light and sound alarm system.

With a smart security camera, you can monitor your home and premise from a smartphone when you are away or on vacation. Smart motion sensors identify the difference between residents, visitors, pets and burglars, and notify you if suspicious behaviour is detected.

Smart toilets. A smart toilet features automatic flushing, automatic cleaning, water overflow protection, tank-leak monitor, health monitoring function to monitor your health by conducting a urine analysis each time you use the toilet and others.

Power consumption by lights and fans can be controlled using an automatic system employing IR or motion sensors. Toilet usage can be tracked using light indicators such as red for occupied and green for vacant.

Sensors can also track bathroom usage, occupancy and stock levels of amenities including toilet papers and soap dispensers. If no one has used the toilet for some time, the cleaning schedule can be adjusted accordingly. Sensors also help you maintain high levels of hygiene.

Smart kitchens. Smart homes have been witnessing innovations in the kitchen for a while now. Modern technologies are spreading to refrigerators, stoves and cooking utensils. Manufacturers are developing synthetic sensor-based devices to monitor kitchen sounds, vibrations, light, heat, gas, electromagnetic noise and temperature. Devices containing sensors are capable of determining precise tasks like checking whether a faucet is running, or a microwave door is open, calculating usage of paper towels from a dispenser and others.

Gardens. You can find many sensors in a garden, including soil moisture, pH, temperature, light level, nutrient and humidity. These allow you to monitor the garden either on a smartphone or a computer.

Beds. A home bed monitoring system equipped with sensors logs your routine, daily activity patterns. This is particularly useful for the elderly or the disabled, living alone. If the activities deviate from the established pattern, the concerned person or caregiver can be notified. Feasibility of incorporating bed sensors as part of a wireless home monitoring system makes the system more flexible and versatile.

Smart thermostats. Integrated with sensors and Wi-Fi, smart thermostats deliver basic control of temperature settings, provide digital interfaces, energy savings, remote programming, geofencing, learning and alerts. It allows you to schedule, monitor and remotely control temperatures and remind you to change filters, among other things.

Smart lightings. You can connect a lamp to a plug, replace hardwired switches, keypads and outlets with wireless sensors and then control these with a remote or smartphone.

In addition to this, smart lighting systems, such as Hue from Philips, can detect occupants in a room and adjust lighting as needed. Smart lightbulbs can regulate energy consumption based on the time of day.

Smart lock systems. Using smart locks and garage door openers, you